

ZIMA, Russian Federation

WMO: 306030

Lat: 53.933N

Lon: 102.050E

Elev: 1503

StdP: 13.92

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-29.1	-39.3	0.6	-34.8	-33.2	0.9	-28.7	17.3	12.7	14.2	10.0	0.2	0	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	84.9	65.1	81.7	63.8	78.7	62.8	68.1	79.1	66.4	77.4	64.9	75.2	4.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	96.0	71.9	62.7	90.0	71.0	61.0	84.8	69.9	33.4	78.9	32.0	77.4	30.9	74.9	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.4	10.6	DB	-36.9	90.8	7.1	3.4	-42.0	93.3	-46.2	95.3	-50.2	97.2	-55.3	99.7	(4)
(5)			WB	-36.9	72.0	6.8	3.5	-41.7	74.6	-45.7	76.6	-49.5	78.6	-54.5	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	-4.9	2.1	18.6	37.1	49.4	61.3	66.0	61.1	48.3	33.6	13.5	-0.2		
	DBStd	26.55	12.55	12.53	11.80	8.59	8.20	6.91	5.11	6.14	7.79	9.16	12.81	13.24		
	HDD50	7820	1702	1341	973	395	114	4	0	4	125	512	1095	1555		
	HDD65	12081	2167	1761	1438	837	486	149	50	149	504	975	1545	2020		
	CDD50	1370	0	0	0	8	96	345	496	350	73	2	0	0		
	CDD65	154	0	0	0	0	3	39	81	30	1	0	0	0		
	CDH74	1939	0	0	0	4	129	595	822	360	29	0	0	0		
	CDH80	470	0	0	0	0	27	164	202	74	3	0	0	0		
(14) Wind	WSAvg	3.7	2.1	2.6	3.8	5.3	5.4	4.4	3.7	3.8	3.8	3.6	3.1	2.3		
(15) Precipitation	PrecAvg	15.2	0.5	0.4	0.4	0.7	1.3	2.3	2.9	2.8	1.9	0.9	0.7	0.7		
	PrecMax	22.8	1.1	0.7	1.0	1.6	3.1	5.6	6.7	6.1	4.8	3.3	1.7	2.0		
	PrecMin	8.3	0.1	0.1	0.0	0.2	0.2	0.7	0.5	1.0	0.7	0.1	0.2	0.1		
	PrecStd	3.5	0.2	0.2	0.2	0.4	0.8	1.1	1.5	1.3	0.9	0.6	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	34.3	48.8	69.6	83.5	89.2	89.2	86.0	78.2	64.0	42.4	30.2		
		MCWB	25.8	28.5	38.2	49.8	58.9	64.9	68.8	65.7	59.9	49.0	36.0	26.9		
	2%	DB	22.3	28.9	43.3	63.1	77.6	84.6	85.0	81.7	72.1	56.5	37.0	25.3		
		MCWB	20.6	25.1	34.9	46.7	56.1	63.6	66.8	64.8	57.1	45.4	32.7	23.5		
	5%	DB	17.2	24.2	39.4	57.5	72.4	80.9	81.8	78.0	67.1	51.2	33.9	21.3		
		MCWB	15.7	21.2	32.7	43.8	53.4	62.4	65.6	63.3	54.6	42.7	31.1	19.9		
	10%	DB	12.7	19.8	35.6	52.3	66.9	76.9	78.7	74.4	62.2	46.7	31.0	17.7		
		MCWB	11.6	17.5	30.1	41.3	50.6	60.9	64.8	61.8	52.2	40.4	28.7	16.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	29.4	39.0	51.3	60.5	68.1	71.6	68.8	61.6	50.1	37.2	27.4		
		MCDB	27.6	33.9	47.0	66.4	79.6	79.8	83.8	80.9	74.2	61.2	41.5	29.6		
	2%	WB	20.9	25.2	35.7	47.8	57.3	66.0	69.1	66.4	58.5	46.5	33.3	23.4		
		MCDB	22.2	28.9	43.1	60.9	74.4	79.2	80.2	77.7	68.9	54.5	35.9	25.1		
	5%	WB	15.8	21.3	33.0	44.8	54.7	64.3	67.4	64.7	55.8	43.4	31.2	19.9		
		MCDB	17.2	24.2	38.7	56.3	69.5	77.5	77.9	75.3	65.1	50.2	33.9	21.2		
	10%	WB	11.6	17.5	30.5	41.7	52.0	62.5	66.0	62.9	53.2	40.8	28.6	16.7		
		MCDB	12.7	19.9	35.4	51.8	65.2	73.6	76.1	72.0	60.4	46.1	31.1	17.7		
(35) Mean Daily Temperature Range	MDBR		16.7	22.0	22.8	19.7	23.3	22.0	19.6	19.8	19.9	17.8	15.7	14.9		
	5% DB	MCDBR	17.3	22.8	23.9	29.9	33.8	30.0	26.8	27.7	28.6	26.2	16.3	15.1		
		MCWBR	15.9	19.0	16.5	16.5	15.8	12.2	11.2	12.5	14.0	15.4	12.7	13.7		
	5% WB	MCDBR	16.9	22.2	21.7	27.5	30.2	24.8	21.6	23.7	25.5	24.2	15.7	15.0		
		MCWBR	15.6	18.5	15.3	16.0	15.0	11.2	10.3	11.5	13.2	14.7	12.6	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.307	0.340	0.385	0.435	0.427	0.423	0.391	0.347	0.327	0.304	0.280		
	taud		2.041	2.028	2.043	2.177	2.088	2.220	2.264	2.336	2.417	2.344	2.154	2.016		
	Ebn at Noon		204	240	260	266	257	261	260	262	261	237	195	174		
	Edh at Noon		23	33	41	41	48	42	40	35	28	24	21	19		
(44) All-Sky Solar Radiation	RadAvg		340	682	1188	1540	1708	1886	1707	1453	1027	618	330	236		
	RadStd		22	38	57	101	141	110	153	89	89	48	29	22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page